SUNSTREAM AVIATION

Clean Fuel, Cleaner Skies

Ethanol to Jet Fuel

Tackling global warming sustainably and economically!

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[1]

AGENDA

SUNSTREAM AVIATION

01 Introduction

How SAF production addresses rising CO_2 levels.

02 Process Description & Equipment List

The ethanol-to-jet process

03 Sample Calculations

Aspen simulation vs. hand calculations

04 Q&A

Time to answer your burning questions!

01

Introduction

What makes SAF production sustainable?

The Global Warming Problem

SUNSTREAM AVIATION

29%

Of all green house gas emissions are due to transportation

Approx. 165 trillion metric tons of CO_2!

9%

Of which is via aircraft!

[1] United States Environmental Protection Agency. *Sources of Greenhouse Gas Emissions*. United States Environmental Protection Agency. https://www.epa.gov/ghgemissions/sources-greenhouse-gas-emissions.

2050

Grand Challenge: Completely replace all Jet A fuel with SAF domestically

Aviation is **exceptionally hard** to decarbonize!
It's mobile, requires high energy density fuel,
and exhibit long lead and service life!

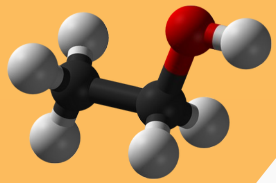
[2] *Sustainable Aviation Fuel Grand Challenge: Tracking Metrics and Mid-2024 Dashboard.* Energy.gov. https://www.energy.gov/eere/bioenergy/articles/sustainable-aviation-fuel-grand-challenge-tracking-metrics-and-mid-2024.

02

Process Description & Equipment List

The ETJ-SAF Pathway.

Chosen Process



Ethanol → Jet fuel

Dehydration → Oligomerization → Hydrogenation → Hydroprocessing

Dehydration

$$CH_3-CH_2-OH \longrightarrow CH_2=CH_2 + H_2O$$

Oligomerization

$$n\ C_2H_4 \longrightarrow C_{2n}H_{4n}$$

Hydrogenation

$$R-CH=CH-R' + H_2 \longrightarrow R-CH_2-CH_2-R'$$

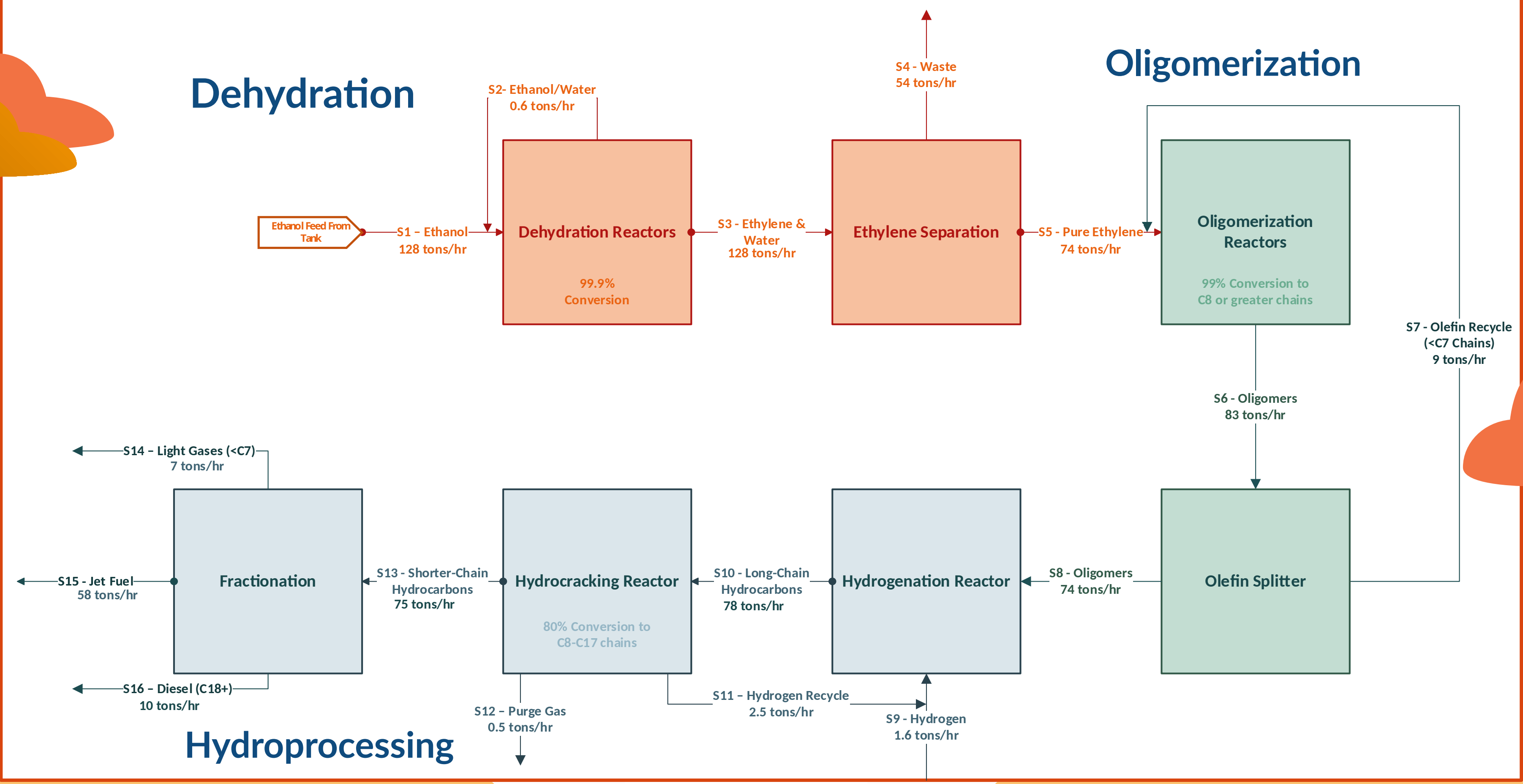
Hydroprocessing

$$R-CH_2-CH_2-R' + H_2 \longrightarrow R-CH_3 + CH_3-R' \text{ (n-parrafins)}$$

$$\text{n-parrafins} \longrightarrow \text{iso-parrafins}$$

[7]

Block Flow Diagram & Material Balances



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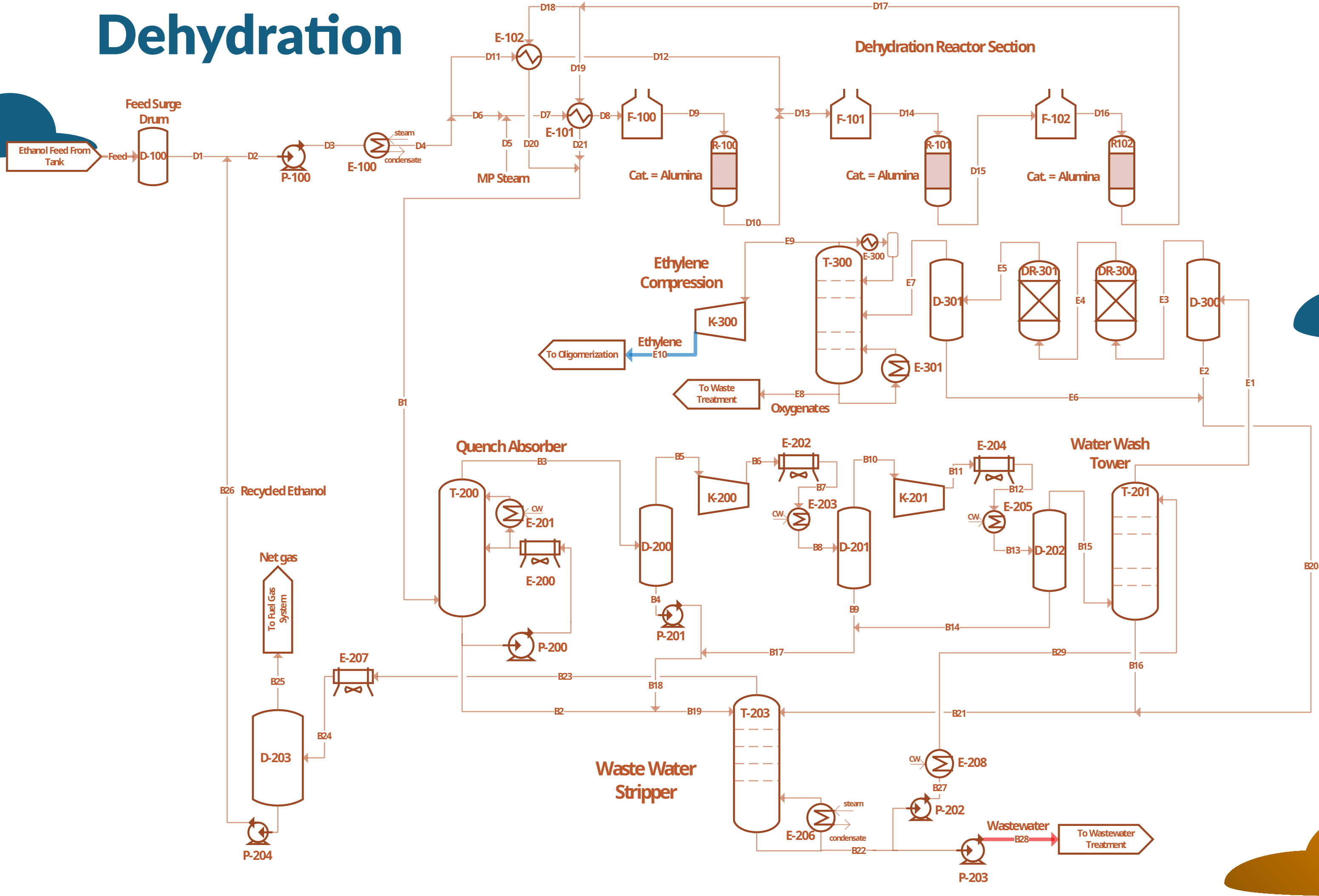
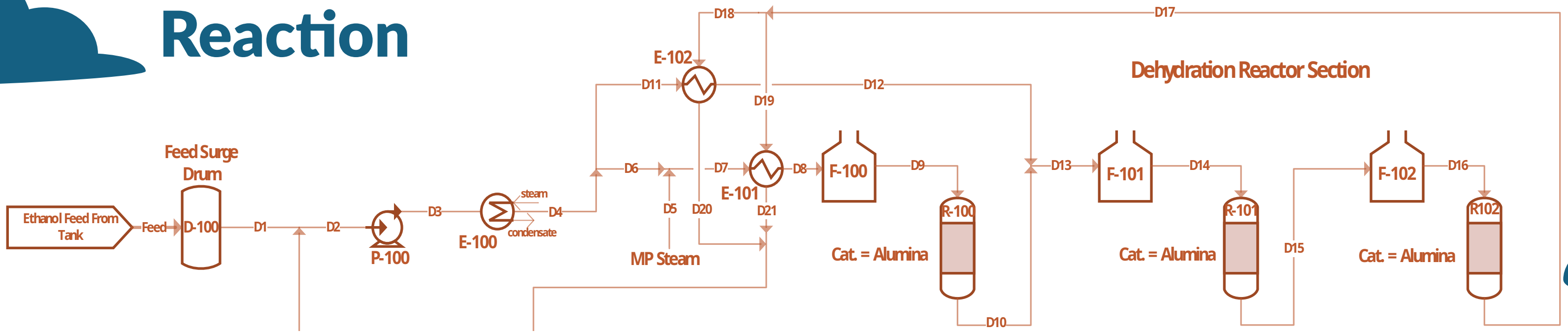
Dehydration

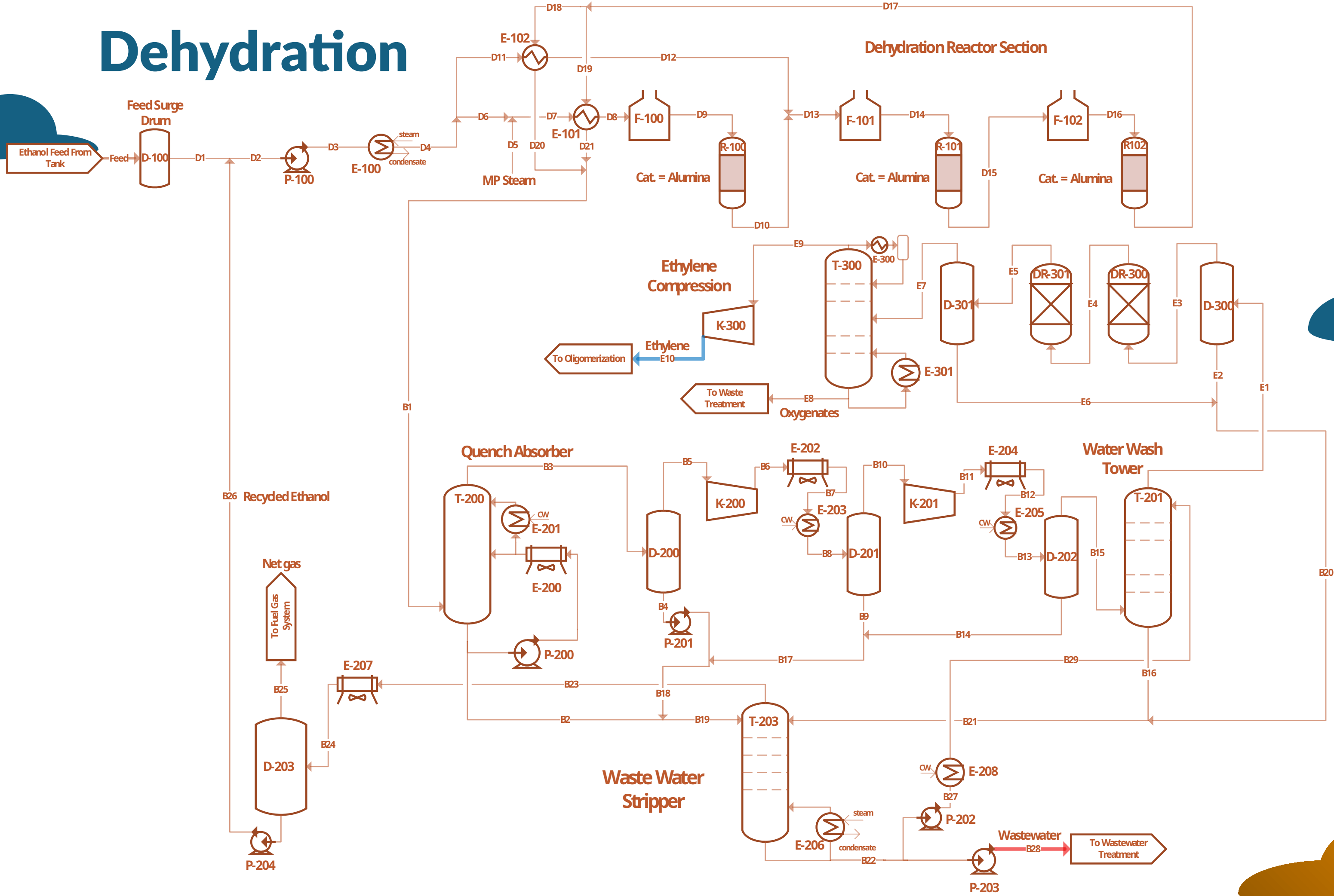


Dehydration Reaction

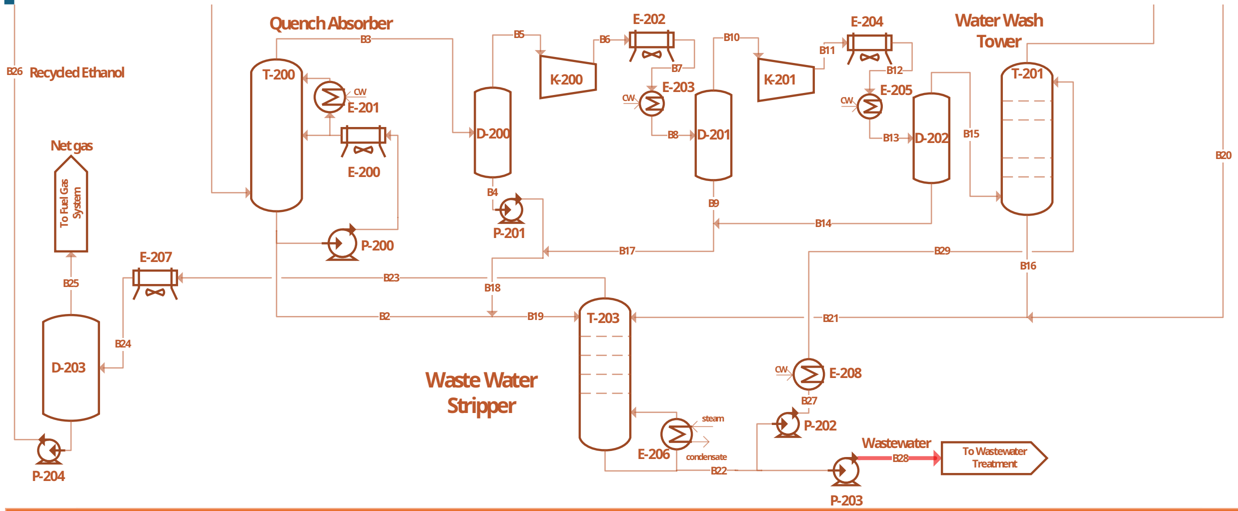


Equipment Name	ID	MOC	Size	Units	Utility/Duty/ Cat.	Units 2	Type
Centrifugal Pump	P-100	Stainless Steel 316	640 30	gal/min hp	0.08	MMBtu/ hr	Electricity
Vaporizer	E-100	Stainless Steel 316	15 129	US Ton MMBtu/ hr	2,480	lb/min	MP Steam
Shell and Tube Heat Exchanger (x2)	E-101	Stainless Steel 316	365	ft²	23	MMBtu/ hr	Process Stream
Fired Heater (x3)	F-100	Stainless Steel 316	503 157	US Ton MMBtu/ hr	10,146	lb/min	Natural Gas
Surge Drum	D-100	Stainless Steel 316	4	US Ton	-	-	-

Dehydration



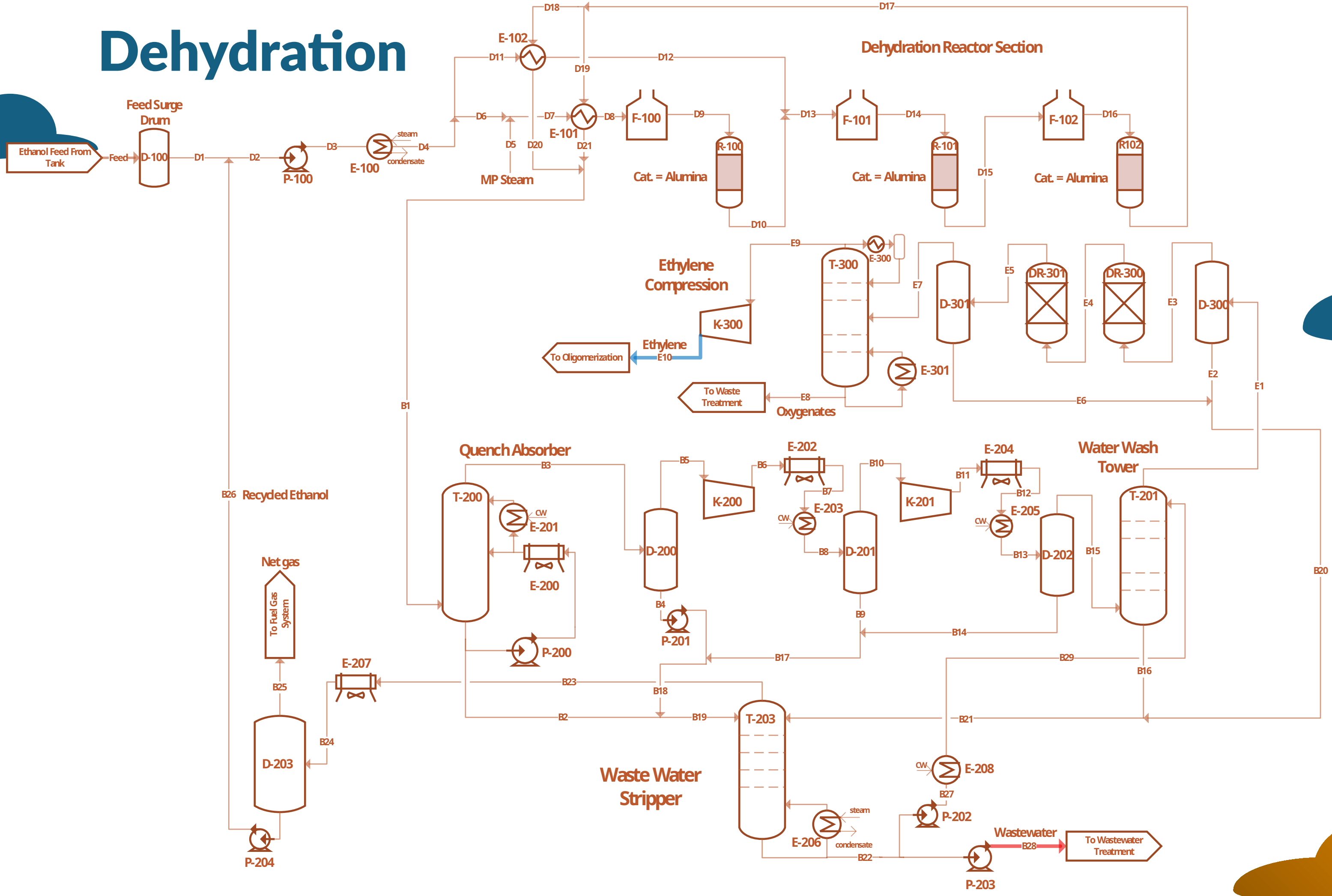
Byproduct Removal



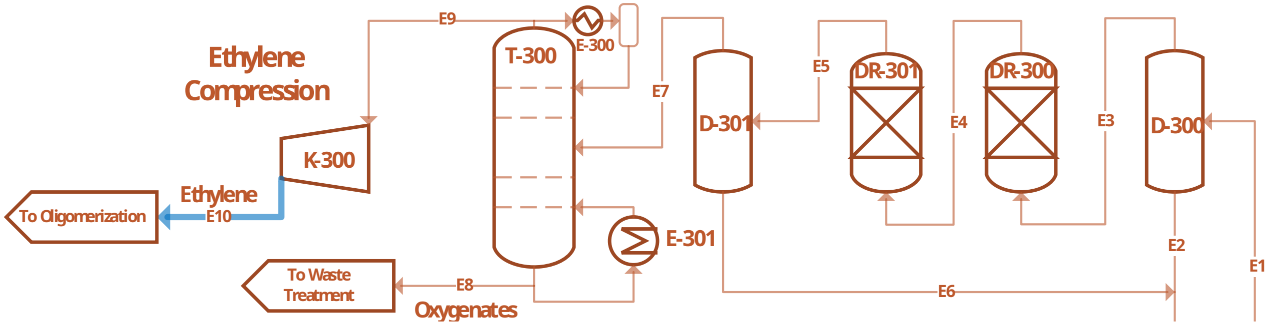
Equipment Name	ID	MOC	Size	Units	Utility/Duty/ Cat.	Units 2	Type
Water Wash Tower (Trayed Column)	T-201	Stainless Steel 316	Diameter: 4.6 Weight: 13	ft US Ton	-	-	-
Sieve Trays	-	Stainless Steel 316	6	stages	-	-	-
Condenser	-	-	-	-	-	-	-
Reboiler	-	-	-	-	-	-	-
Water Stripper (Trayed Column)	T-202	Stainless Steel 316	Diameter: 6.9 Weight: 12	ft US Ton	-	-	-
Sieve Trays	-	Stainless Steel 316	9	stages	-	-	-
Shell & Tube Condenser	-	Stainless Steel 316	411	ft^2	10 1,035	MMBtu/hr gal/min	Cooling Water
Thermosiphon	E-206	Stainless Steel	4,052	ft^2	52	MMBtu/hr	LP Steam

Compon	B29
Temperat	275
Pressu	411
Mass Flo	11,023
Ethan	0
Water	11,018

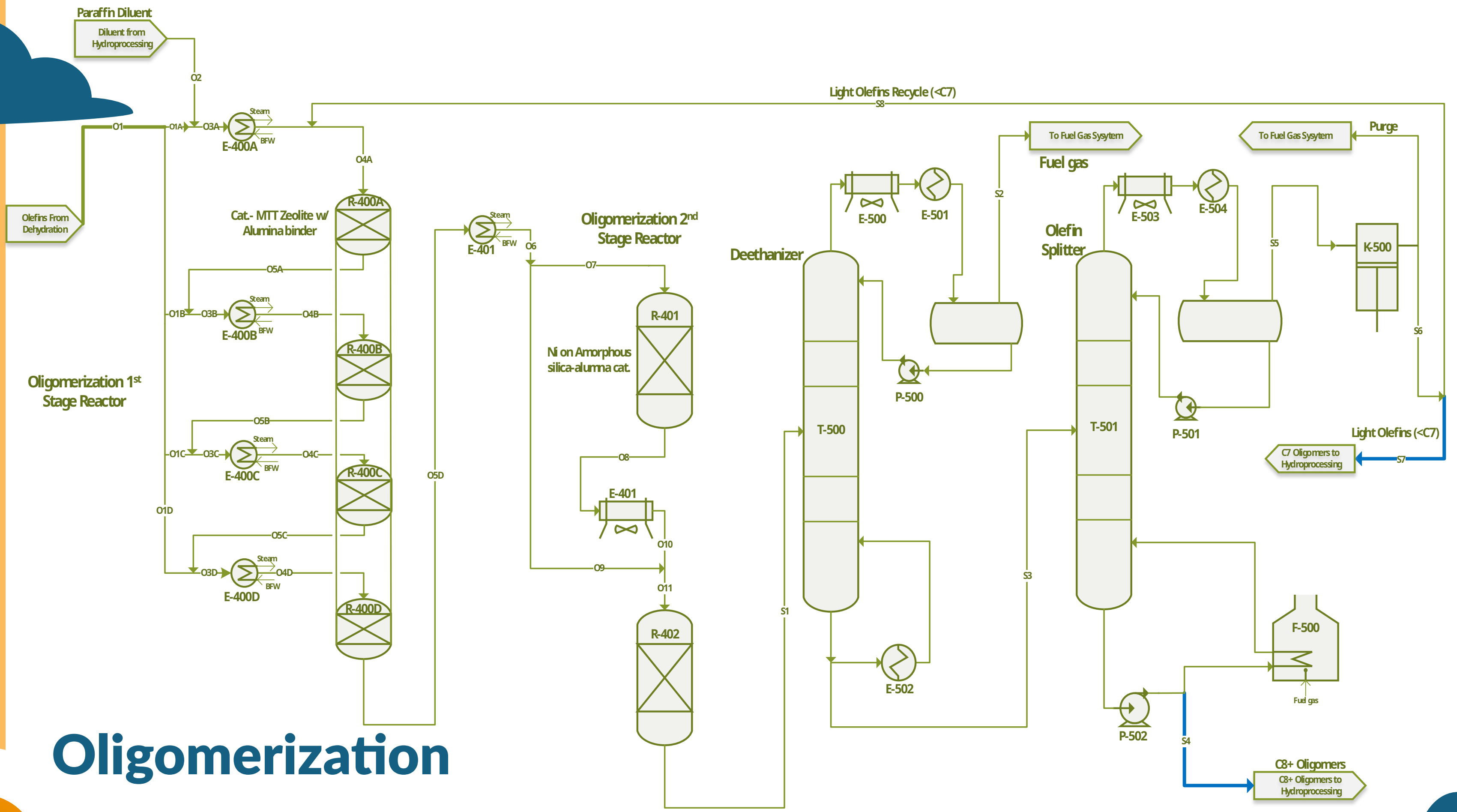
Dehydration



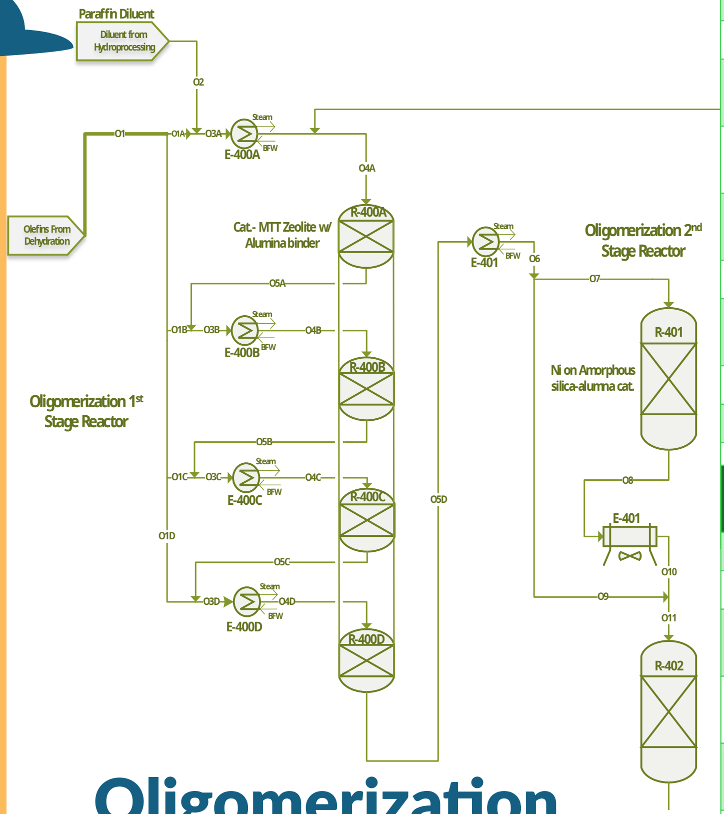
Ethylene Purification



Equipment Name	ID	MOC	Size	Units	Utility/Duty/Cat.	Units 2	Type
Centrifugal Compressor	K-300	Carbon Steel	2,299	hp	6	MMBtu/hr	Electricity
Knockout Drum (x2)	D-300	Stainless Steel 316	10	US Ton	-	-	-
Adsorbent Drier (x2)	DR-300	Stainless Steel 316	-	-	732	ft³	Adsorbent
Oxygenate Stripper (Trayed Column)	T-300	Carbon Steel	Diameter: 7 Weight: 77	ft US Ton	-	-	-
Sieve Trays	-	Carbon Steel	26	stages	-	-	-
Shell & Tube Condenser	E-300	Carbon Steel	12,347	ft²	9 1,221	MMBtu/hr lb/min	Refrigerant
Thermosiphon					5	MMBtu/	



Oligomerization

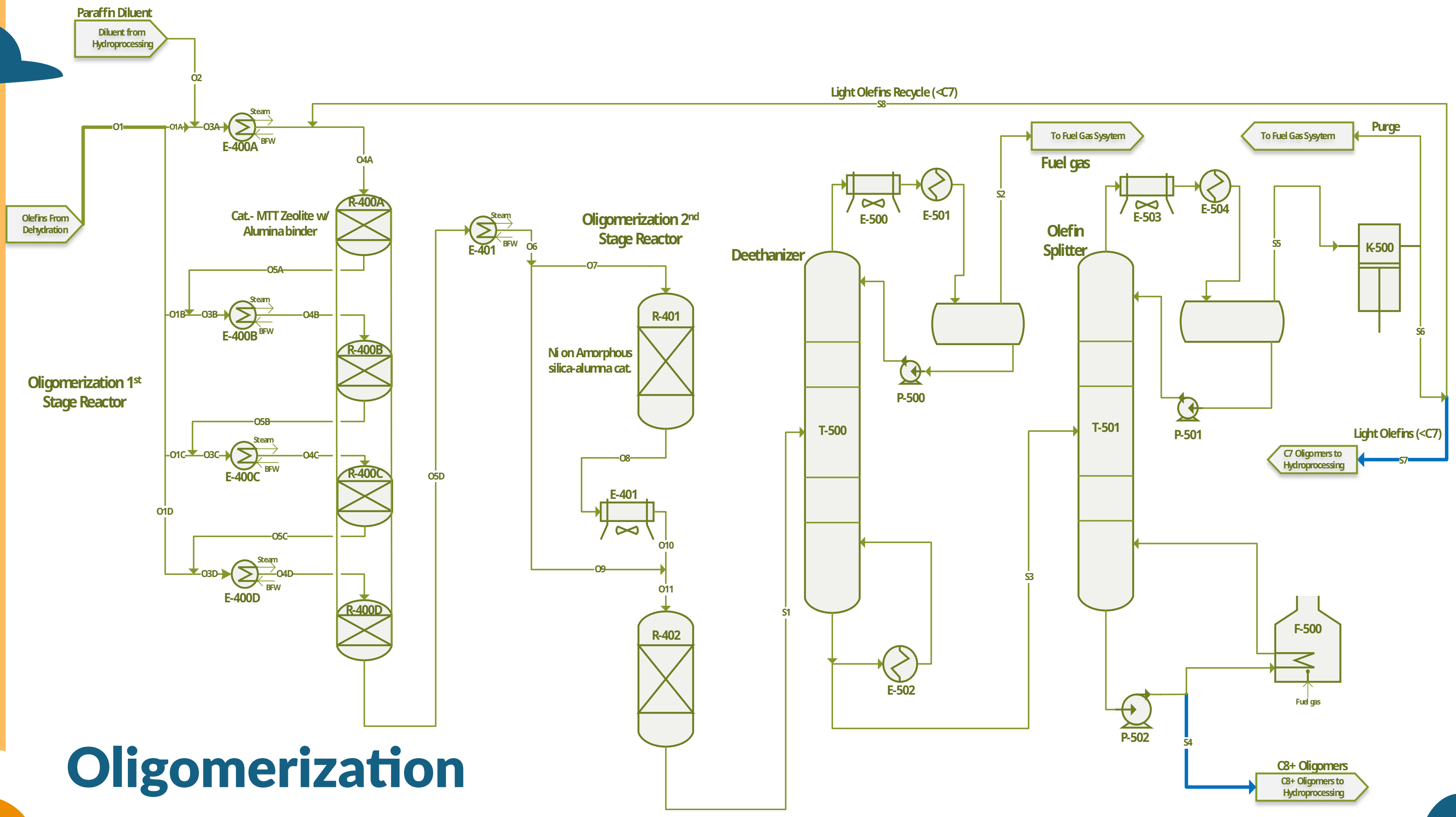


Oligomerization Reaction

Component	Units	O1	O1A	O2	O3A	O4A	O5A	O1B	O3B	O4B
Temperature	F	112	198	410	265	**248**	**320**	198	283	**248**
Pressure	psia	1,015	1,005	1,005	1,005	1,000	1,010	1,005	1,005	1,000
		147,69	46,14				64,45		110,59	110,59
Pentene	lb/hr	0	0	0	0	0	0	0	0	**0**
Hexene	lb/hr	21,806	0	21,806	21,806	30,567	0	30,567	30,567	**39,329**
Heptene	lb/hr	0	0	0	0	0	0	0	0	**0**
Octene	lb/hr	2,726	0	2,726	2,726	3,821	0	3,821	3,821	**4,916**

Component	Units	O6	O7	O8	O9	O10	O11	S1
Temperature	F	248	**248**	**320**	248	248	**248**	**320**
Pressure	psia	990	990	990	990	985	985	**985**
Mass Flows	lb/hr	202.875	101.438	101.438	101.438	101.438	202.875	**202.875**
C_2-C_7 Olefins	lb/hr	160.608	80.304	18.208	80.304	18.208	98.512	**36.415**
C_8-C_{17} Olefins	lb/hr	4.916	2.458	62.899	2.458	63.727	65.357	**125.798**
C_{18+} Olefins	lb/hr	-	-	1.655	-	-	1.655	**3.310**
Parraffins	lb/hr	36.874	18.437	18.437	18.437	18.437	36.874	**36.874**
Non-Condensables	lb/hr	198	98	98	98	98	198	**198**

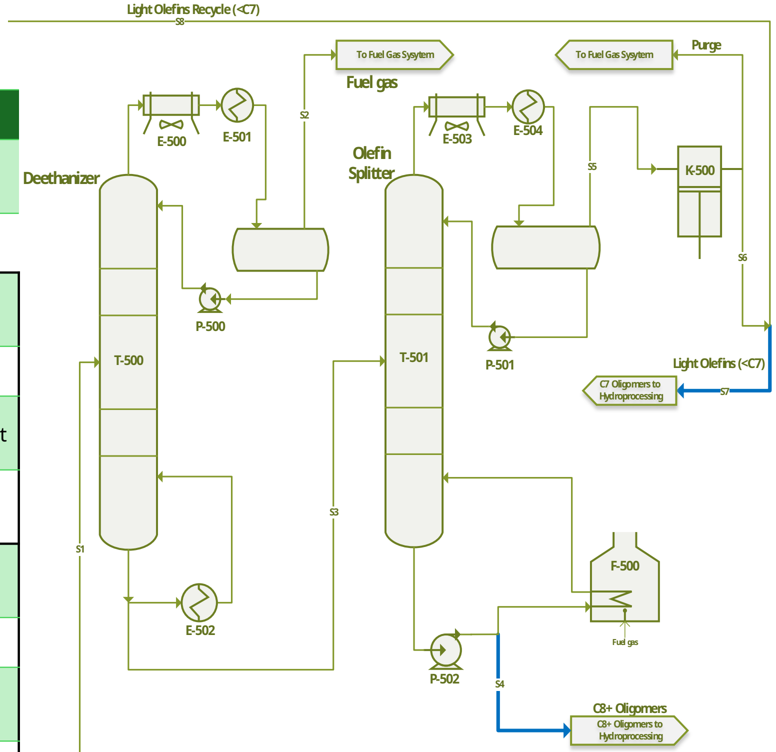
Air Cooler	E-405	Carbon Steel	3,015	ft²	5	hr	Air
Fixed Bed Reactor (x4)	R-400	Carbon Steel	114	US Ton	664	ft³	MTT Zeolite on Alumina Catalyst
Fixed Bed Reactor (x2)	R-404	Carbon Steel	53	US Ton	554	ft³	Amorphous Alumina Catalyst



Oligomerization

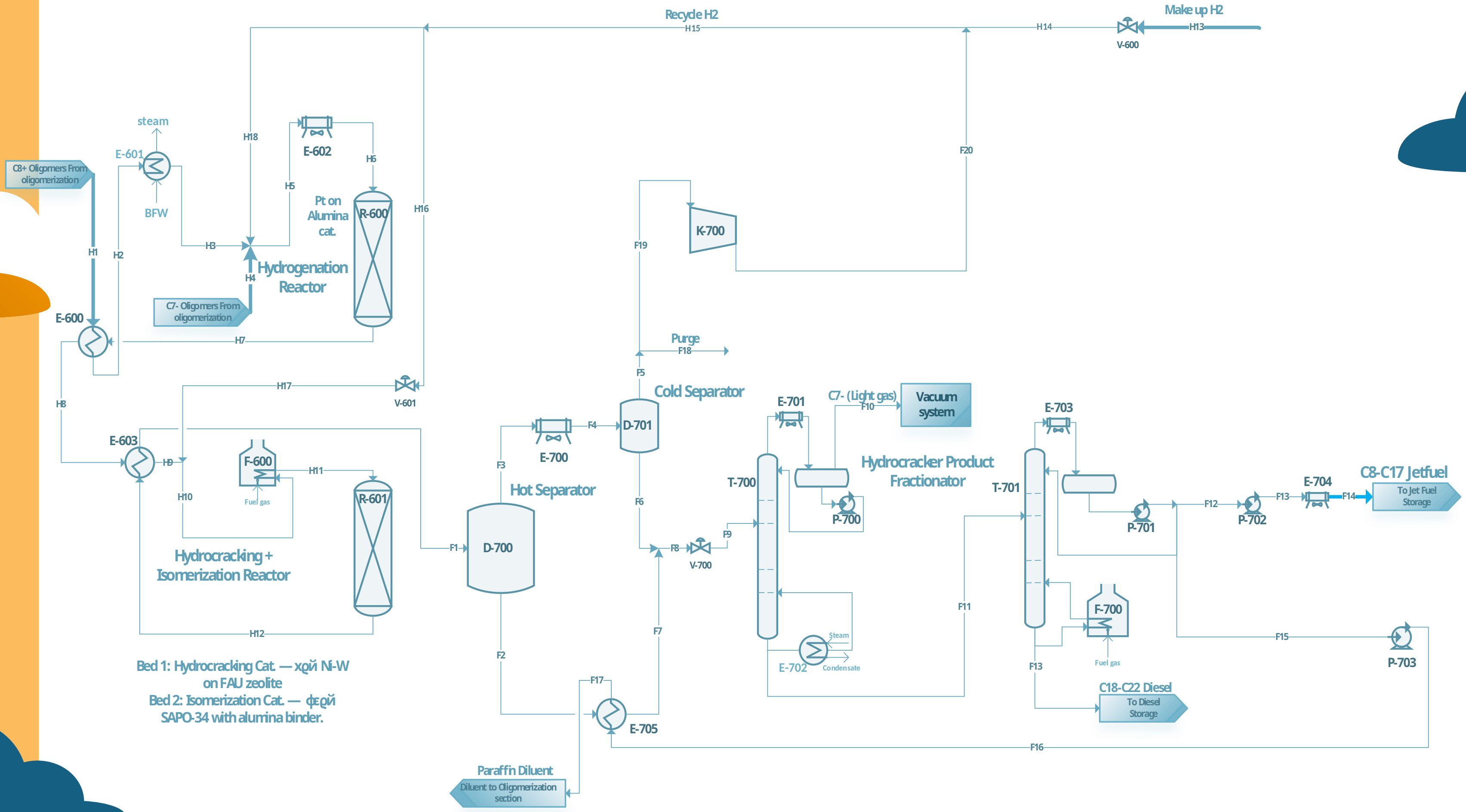
Equipment Name	ID	MOC	Size	Units	Utility/Duty/ Cat.	Units 2	Type
Centrifugal Pump	P-502	Carbon Steel	561 377	gal/ min hp	1	MMBtu/ hr	Electricity
Centrifugal Compressor	K-500	Carbon Steel	782	hp	2	MMBtu/ hr	Electricity
Deethanizer (Trayed Column)	T-500	Carbon Steel	Diameter: 12.3 Weight: 23	ft US Ton	-	-	-
Sieve Trays	-	Carbon Steel	25	stages	-	-	-
Shell & Tube Condenser	E-501	Carbon Steel	269	ft^2	0.7 87	MMBtu/ hr lb/min	Refrigerant
Thermosiphon Reboiler	E-502	Carbon Steel	676	ft^2	6 132	MMBtu/ hr lb/min	HP Steam
Olefin Splitter (Trayed Column)	T-501	Carbon Steel	Diameter: 11.4 Weight: 120	ft US Ton	-	-	-
Sieve Trays	-	Carbon Steel	33	stages	-	-	-
Shell & Tube Condenser	E-504	Carbon Steel	1,672	ft^2	38 35,587	MMBtu/ hr gal/min	Cooling Water
Reboiler	F-500	Carbon Steel	156	US Ton	42 2,708	MMBtu/ hr lb/min	Natural Gas

Component	Unit	S1	S2	S3	S4
C_8-C_{17} Olefins	lb/hr	1	1	0	0
C_{18+} Olefins	lb/hr	0	0	0	0

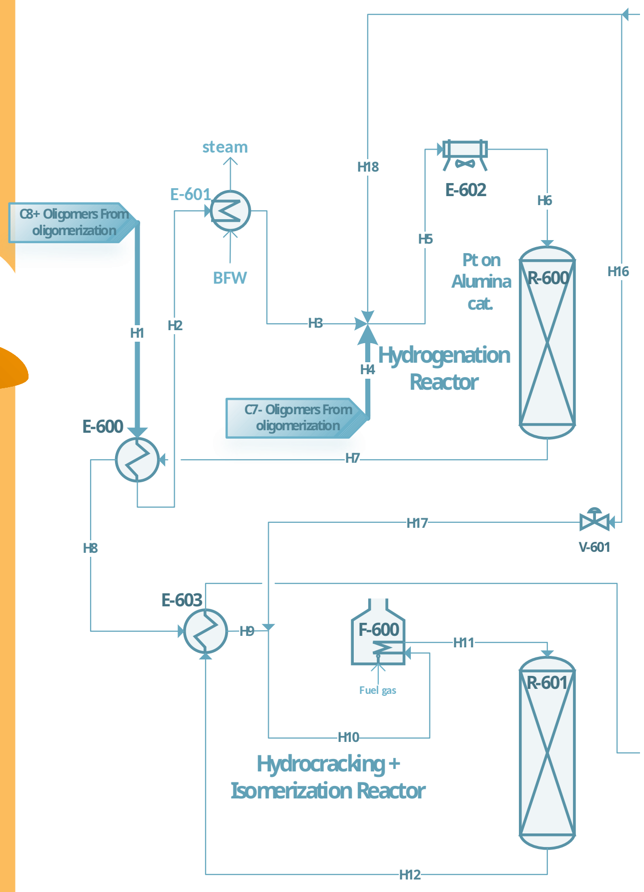


Olefin Separation

Hydroprocessing



Hydroprocessing Reactions

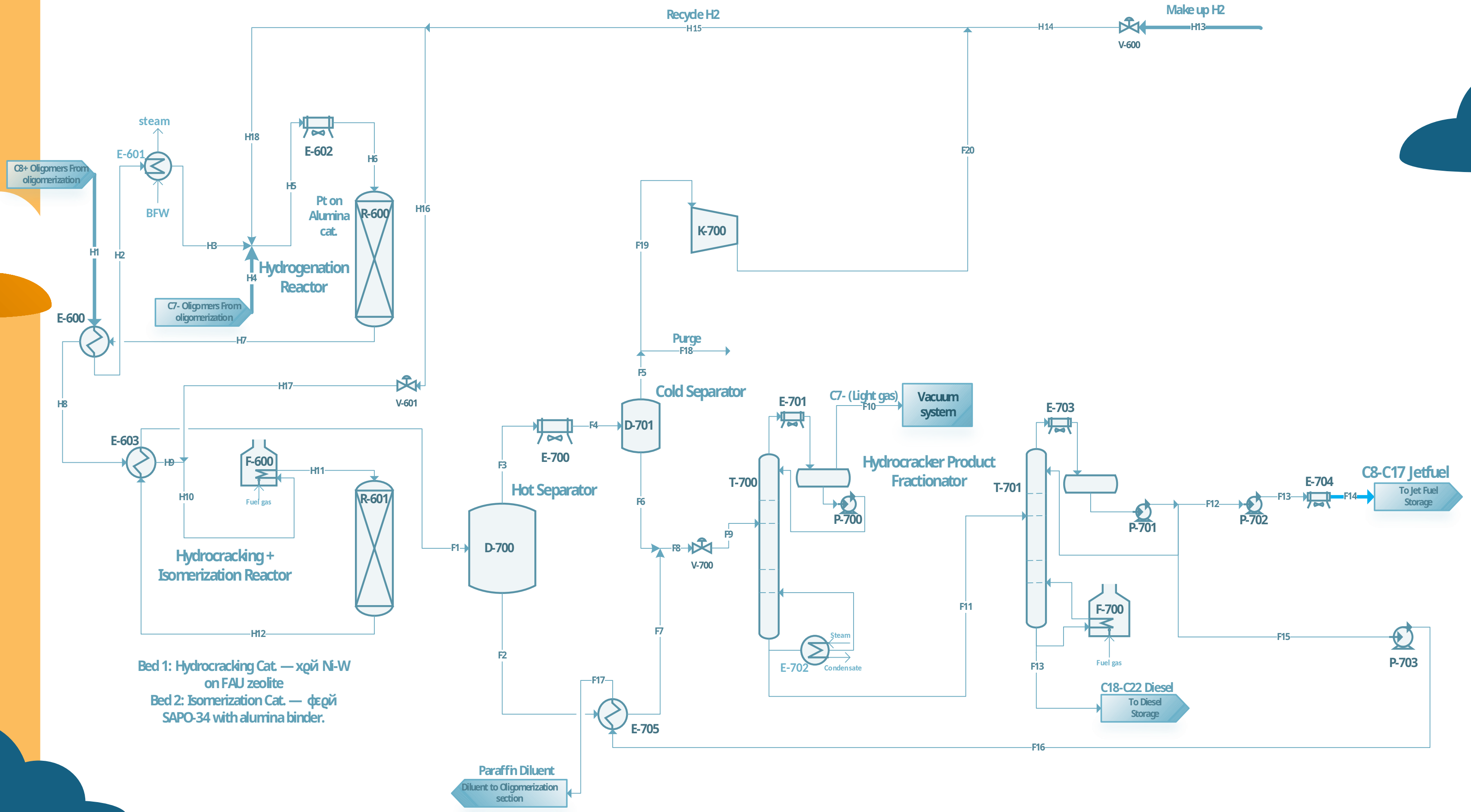


Bed 1: Hydrocracking Cat. —xọй Ni-W on FAU zeolite
Bed 2: Isomerization Cat. — фεọй SAPO-34 with alumina binder.

Component	Units	H9	H10	H11	H12	H13
Temperature	F	716	685	752	748	257
Pressure	psia	909	904	899	834	2.176
Mass Flows	lb/hr	192.325	195.769	195.769	195.769	3.107
Hydrogen	lb/hr	3.484	5.927	5.927	5.605	3.107
Hydrogen	lb/hr	3.107	8.144	2.443	2.443	5.701
C_2-C_7 Olefins	lb/hr	-	-	-	-	-
C_8-C_{17} Olefins	lb/hr	-	-	-	-	-
C_{18+} Olefins	lb/hr	-	-	-	-	-

Equipment Name	ID	MOC	Size	Units	Utility/Duty/Cat.	Units 2	Type
Shell and Tube Heat Exchanger (x2)	E-600	Stainless Steel 316	5,107	ft^2	41	MMBtu/hr	Process Stream
Air Cooler	E-603	Stainless Steel 316	1,301	ft^2	3	MMBtu/hr	Air
Steam Generator	E-602	Carbon Steel w/ Stainless Internals	498	ft^2	17 1,861	MMBtu/hr gal/min	Process Water
Fired Heater	F-600	Stainless Steel 316	79 12	US Ton MMBtu/hr	339	kg/min	Natural Gas
Hydrogenation Reactor	R-600	Stainless Steel 316	35	US Ton	1,187	ft^3	Platinum on Alumina Catalyst
Hydrocracking & Isomerization Reactor	R-601	Stainless Steel 316	15	US Ton	5,066	ft^3	Ni-W on FAU Zeolite Catalyst

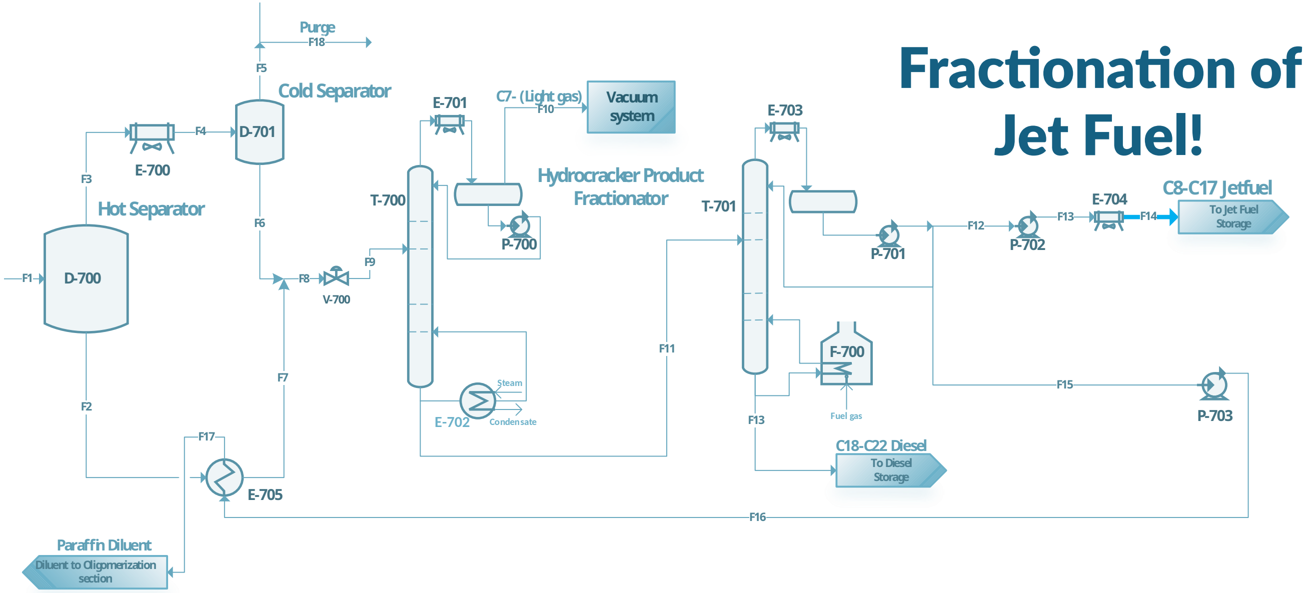
Hydroprocessing



Bed 1: Hydrocracking Cat. — xọй Ni-W on FAU zeolite
Bed 2: Isomerization Cat. — φεọй SAPO-34 with alumina binder.

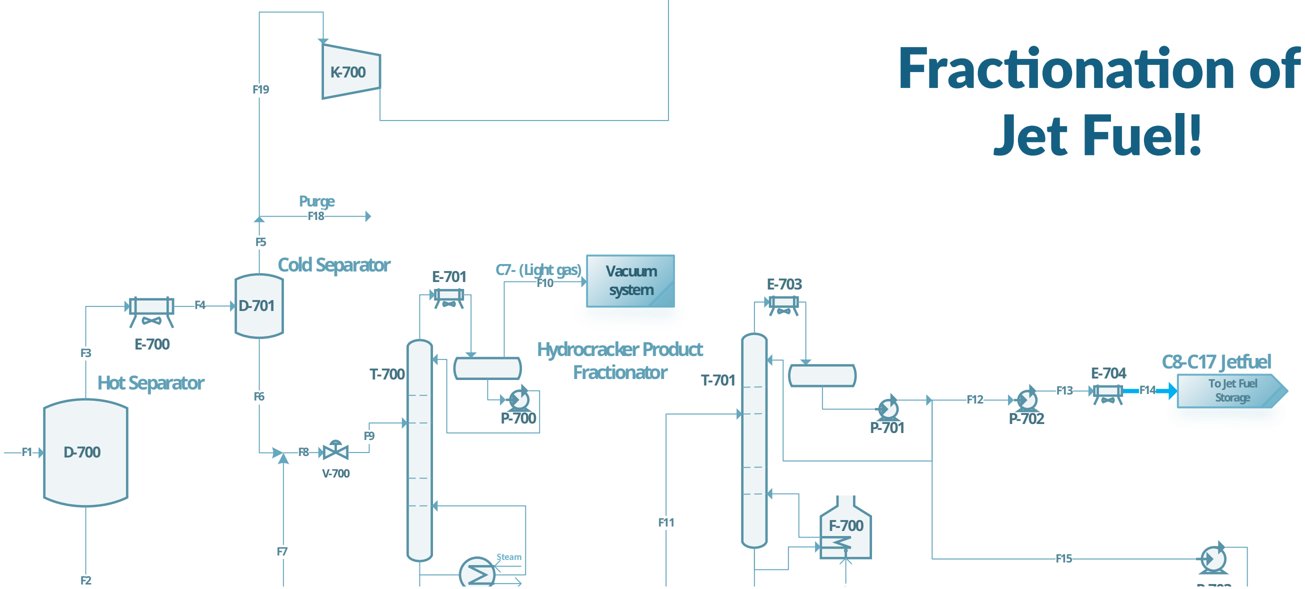
[21]

Fractionation of Jet Fuel!



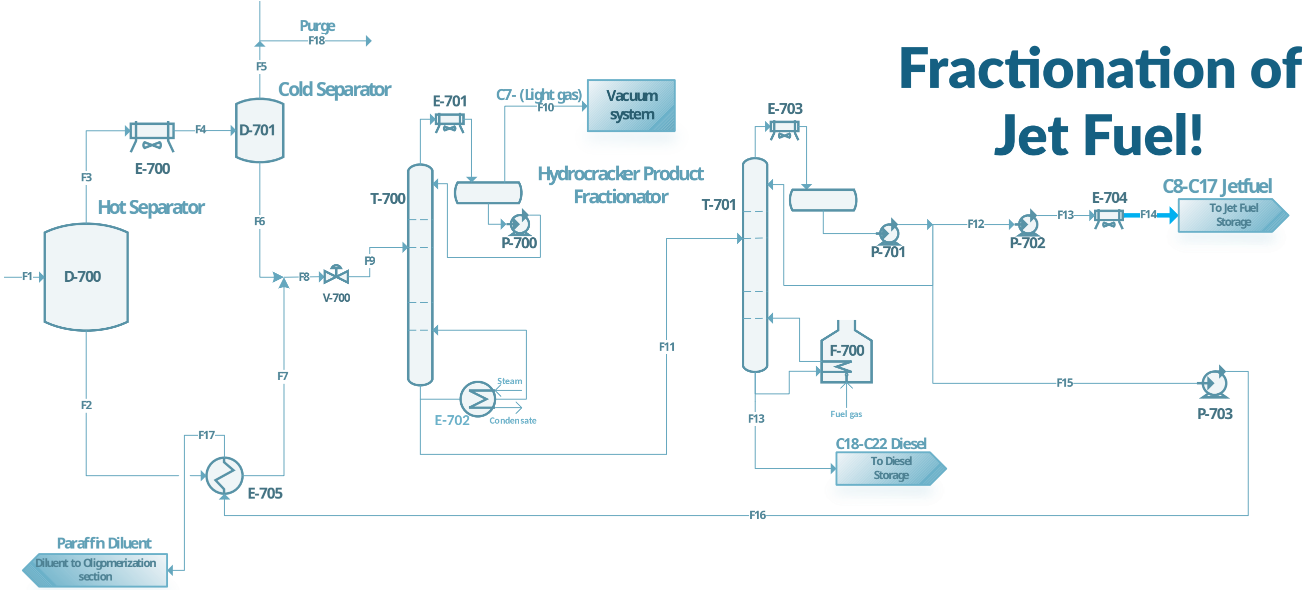
Component	Units	F1	F2	F3	F4	F5	F6	F7	F8	F9
Temperature	F	636	636	636	176	176	176	610	404	391
Pressure	psia	824	824	824	819	819	819	814	814	44
Mass Flows	lb/hr	195,769	91,693	104,076	104,076	9,306	94,770	91,693	186,463	186,463
Hydrogen	lb/hr	5,605	3	5,602	5,602	5,596	5	3	8	8
C_2-C_7 Olefins	lb/hr	-	-	-	-	-	-	-	-	-
C_8-C_{17} Olefins	lb/hr	-	-	-	-	-	-	-	-	-
C_{18+} Olefins										

Fractionation of Jet Fuel!



Component	Units	F1	F2	F3	F4	F5	F6	F7	F8	F9
Temperature	F	636	636	636	176	176	176	610	404	391
Pressure	psia	824	824	824	819	819	819	814	814	44
Mass Flows	lb/hr	195,769	91,693	104,076	104,076	9,306	94,770	91,693	186,463	186,463
Hydrogen	lb/hr	5,605	3	5,602	5,602	5,596	5	3	8	8
C_2-C_7 Olefins	lb/hr	-	-	-	-	-	-	-	-	-
C_8-C_{17} Olefins	lb/hr	-	-	-	-	-	-	-	-	-
C_{18+} Olefins										

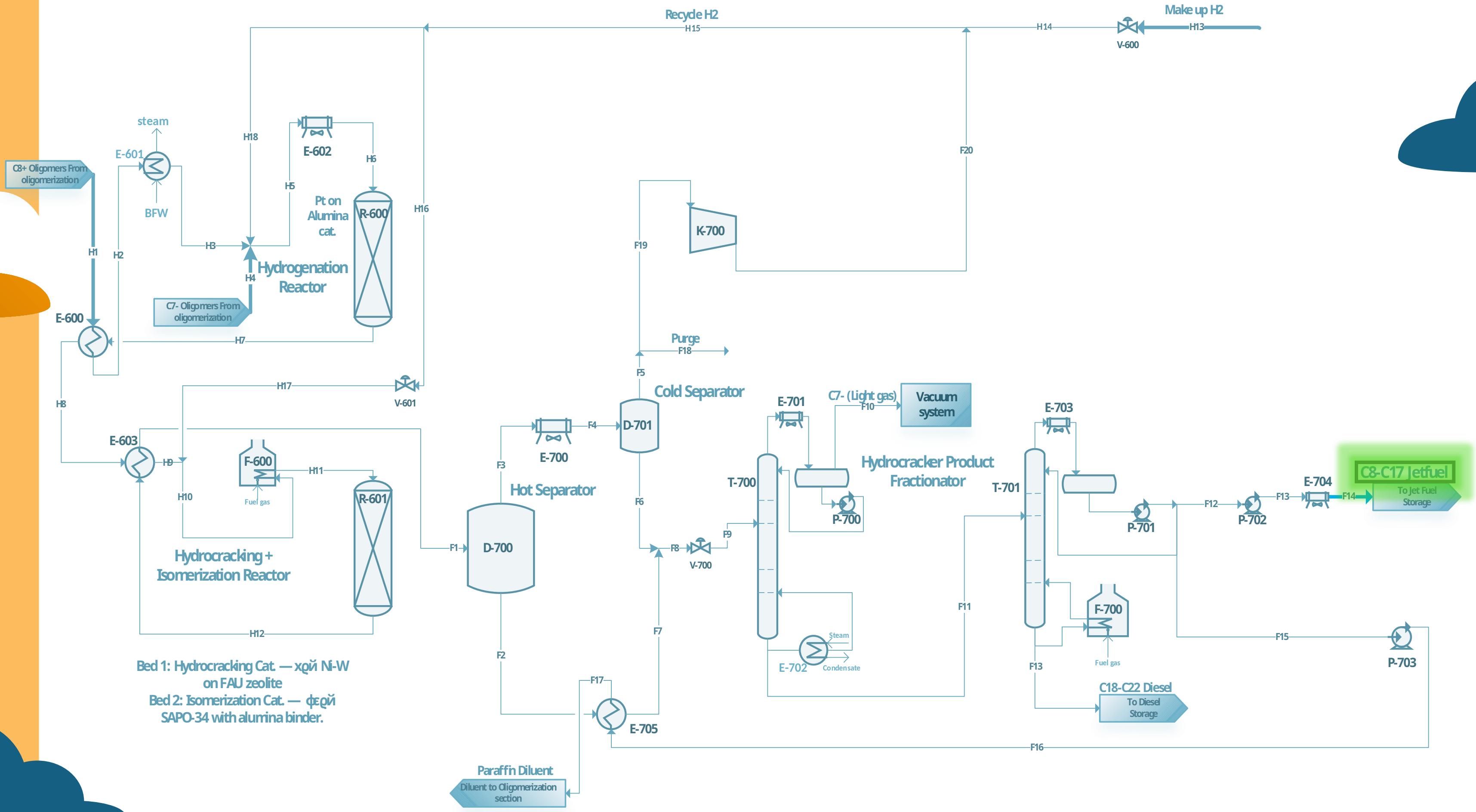
Fractionation of Jet Fuel!



Equipment Name	ID	MOC	Size	Units	Utility/ Duty/Cat.	Units 2	Type
Naptha Fractionator (Trayed Column)	T-700	Stainless Steel 316	Diameter: 6 Weight: 69	ft US Ton	-	-	-
Sieve Trays	-	Stainless Steel 316	31	stages	-	-	-
Condenser	E-701	Stainless Steel 316	13,264	ft²	10	MMBtu/ hr	Air
Thermosiphon Reboiler	E-702	Stainless Steel 316	508	ft²	5 120	MMBtu/ hr lb/min	HP Steam
Jet Fuel Fractionator (Trayed Column)	T-701	Stainless Steel 316	Diameter: 14 Weight: 300	ft US Ton	-	-	-
Sieve Trays	-	Stainless Steel 316	60	stages	-	-	-
[illegible]	[illegible]	Stainless Steel	[illegible]	[illegible]	[illegible]	MMBtu/	[illegible]

Equipment N…	…	Type
Centrifugal Pu… (x2)	…tu/	Electricity
Centrifuga… Compresso…	…tu/	Electricity
Shell and Tube Exchange…	…tu/	Process Stream
Air Cooler (x2)	…tu/	Air
Hot Separat…		

Hydroprocessing



Total Utilities

Dehydration

Utility	Heat/ Energy	Units	Flowrate	Units2
LP Steam	57	MMBtu/ hr	1,094	lb/min
MP Steam	129	MMBtu/ hr	2,480	lb/min
HP Steam	0	MMBtu/ hr	0	lb/min
Natural Gas	157	MMBtu/ hr	10,146	lb/min
Cooling Water	40	MMBtu/ hr	4,648	gal/min

Hydroprocessing

Utility	Heat/ Energy	Units	Flowrate	Units2
LP Steam	0	MMBtu/ hr	0	lb/min
MP Steam	0	MMBtu/ hr	0	lb/min
HP Steam	5.3	MMBtu/ hr	120	lb/min
Natural Gas	123	MMBtu/ hr	3,175	lb/min

Utility	Heat/ Energy	Units	Flowrate	Units2
LP Steam	0	MMBtu/ hr	0	lb/min
MP Steam	0	MMBtu/ hr	0	lb/min
Electricity	26			
HP Steam	6	MMBtu/ hr	132	lb/min
Natural Gas	42	MMBtu/ hr	2,708	lb/min

Total

Utility	Heat/ Energy	Units	Flowrate	Units2
LP Steam	57	MMBtu/ hr	1,094	lb/min
MP Steam	129	MMBtu/ hr	2480	lb/min
Air	122			
HP Steam	11	MMBtu/ hr	252	lb/min
Natural Gas	322	MMBtu/ hr	16,029	lb/min

[26]

03

Sample Calculations

Does our simulation match reality?

Olefin Splitter Calculations

1. Fenske Equation – Minimum Stages

$$N_{min} = \frac{ln\left[\left(\frac{x_{D,LK}}{x_{D,HK}}\right)\left(,\frac{x_{B,HK}}{x_{B,LK}}\right)\right]}{\ln(\alpha_{LK,HK})}$$

$$\boldsymbol{N_{min} = 21.2\ stages}$$

2. Gilliland & Underwood Equations - Stages

$$X = \frac{(R - R_{min})}{(R + 1)} \qquad Y = \frac{N_{eq} - N_{min}}{(N_{eq} + 1)}$$

$$R_{min} = \frac{x_{D,i}\alpha_i}{\alpha_i - \theta} - 1 \qquad \boldsymbol{N_{eq} = 28.2}$$

$$\boldsymbol{R_{min} = 3.49}$$

3. Tray Efficiency (E = 0.80)

$$N_{actual} = \frac{N_{eq}}{E}$$

$$\boldsymbol{N_{actual} = 35\ stages}$$

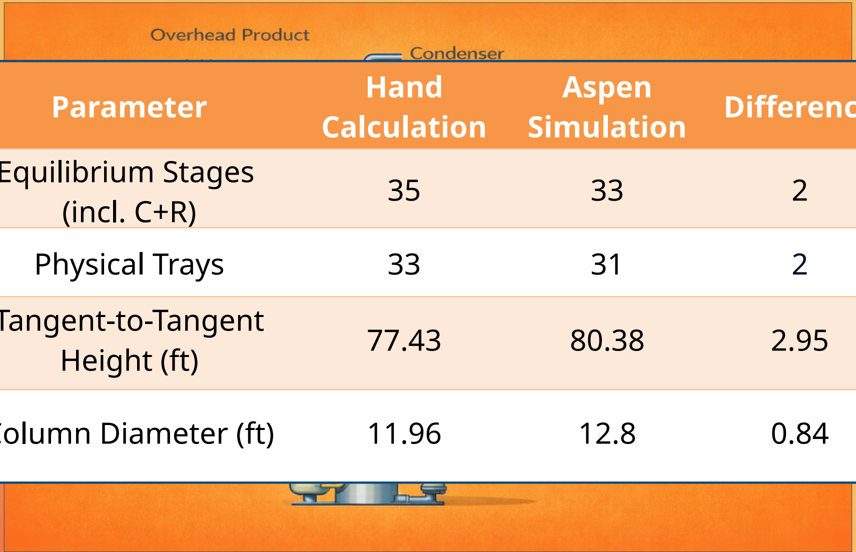
4. Dimensions

$$\mathrm{Height} = \left(\mathrm{N_{trays}} * \mathrm{S}\right) + \mathrm{H_{top}} + \mathrm{H_{bot}}$$

$$\mathbf{Height = = 77.43\ ft}$$

$$Diameter = \sqrt{\frac{4A}{\pi}}$$

$$\boldsymbol{Diameter = 11.96\ ft}$$



Parameter	Hand Calculation	Aspen Simulation	Difference
Equilibrium Stages (incl. C+R)	35	33	2
Physical Trays	33	31	2
Tangent-to-Tangent Height (ft)	77.43	80.38	2.95
Column Diameter (ft)	11.96	12.8	0.84

SPECIAL THANKS

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LANCE BAIRD

Mentor

BETUL BILGIN

Advisor

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ANY QUESTIONS?